Samriddha Ganguly

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Bio-Sketch

I am a fourth-year BS–MS Dual Degree student in Physics at the Indian Institute of Science Education and Research (IISER) Bhopal. My research interests lie in theoretical condensed matter physics, particularly in **quantum materials** and their emergent phenomena.
I am especially interested in exploring **altermagnetic systems** and the **emergence of superconducting states in altermagnets**. My broader goal is to investigate how quantum magnetic systems and topological phases of matter can be engineered for robust quantum information processing using **symmetry-based studies.**
I am particularly motivated by the interplay between **theoretical modeling and computational innovation** in advancing our understanding of correlated quantum matter and how to realize them in labs.

Education

BS-MS IISER Bhopal, Physics — Dec 2022 – Present

- GPA: 3.8/4.0
- **Relevant Coursework:** Introduction to Quantum Physics, Quantum Mechanics I, Advanced Quantum Mechanics, Statistical Mechanics, Numerical Methods in Programming, Wave and Optics, Electromagnetism, Classical Mechanics, General Properties of Matter, Physics through Computational Thinking, Classical Thermodynamics, Basic Electronics, Linear Algebra, Complex Variables, Probability and Statistics, Multivariable Calculus, Groups and Symmetry, Real Analysis, Mathematical Methods for Physicists

BS-MS St. Xavier's Institution, Higher Secondary — May 2020 – Apr 2022

- 98.75%

BS-MS St. Xavier's Institution, Secondary — June 2011 – Apr 2020

- 98%

Experience

Research Intern, Indian Institute of Science Education and Research, Bhopal — Bhopal, India, Jun 2025 – Oct 2025

- Studying about classification of topological quantum matter with symmetries under Dr. Nirmal Ganguli.
- Specifically, I carried out a comprehensive theoretical analysis of the symmetry-protected **topological Dirac nodal line (DNL) magnonic phase** in layered honeycomb collinear antiferromagnets and how it is affected by Interlayer Spin Canting Interactions.
- Work related to this work has been accepted for Poster Presentation in 12th National Conference on Recent Trends in Material Science and Technology, IIST Trivandrum and we are working on a paper submission.

TCS Research and Innovation Fellow TCS Innovation Labs, IIT Kharagpur Research Park. — Kolkata, India, May 2025 – Jul 2025

- I worked in the **Acoustic and Quantum Sensing Lab** on some interesting Industry-related problems. This experience helped me gain some valuable industrial R & D exposure on optimizing workings of Mills and Pipelines.

INSA-IASc-NASI Summer Research Fellow, Indian Institute of Science — Bangalore, India, May 2024 – Jul 2024

- Selected in prestigious Summer Research Fellowship program to gain research experience working under **Dr. Arvind Ayyer, Department of Mathematics, IISC, Bengaluru**.
- This project aims to give a theoretical background to **Markov Processes** with particular emphasis on using MCMC to simulate the Ising Model in 1-D.

RAWS Winter Scholar, IUCAA-NCRA TIFR — Pune, India, Dec 2023 – Dec 2023

- Attended the **Radio Astronomy Winter School (RAWS) 2023** from December 12 to Decem-

ber 22 organized by NCRA-TIFR and IUCAA, Pune. Learned about the **basics of Radio Astronomy**, with **Hands-on Experiments** using Radio Observation Equipments.

UC Berkeley REYES Research Fellow, Temple University — Remote, Jul 2024 – Aug 2024

- Selected to participate in the **UC Berkeley REYES Program** where I worked under the guidance of **Dr. Martha Constantinou from Dept. of Physics, Temple University**.
- I worked on the project titled **"Internal structure of hadrons from numerical simulations of Quantum Chromodynamics"**.

Awards

2025	**TCS Research and Innovation Fellowship:**	50000/-
2023	**CNR Rao Education Foundation Prize:** *Awarded by the Hon'ble Director of IISER Bhopal for having the Highest CPI in the First Semester of First Year of degree.*	5000/-
2022	**Academic Excellence in Higher Secondary Examination:** *Received award from Hon'ble Chief Minister of West Bengal for achieving AIR 5 in HS Examination.*	N/A
2022	**DST INSPIRE FELLOW**	300,000/-
2020	**JBNSTS Junior Fellow:**	40000/-
2020	**INSA-IASc-NASI Summer Research Fellowship:**	25000/-

Initiatives and Projects

open_board: An Interdisciplinary Knowledge-Sharing Initiative — Website | Repo
Co-Founder & Lead Developer

- Conceptualized and developed a peer-to-peer platform for students to present projects and research to a non-specialist, interdisciplinary audience in fun and engaging live sessions.

A Hitchhiker's Guide to Physics @ IISERB — Website | Repo
Lead Developer

- Built a comprehensive website for the physics department to centralize academic resources, featuring course guides and timetable and a platform for sharing notes and a secure, domain-restricted (`@iiserb.ac.in`) forum for discussions.

Skills

Programming and Software: Python • MATLAB • Mathematica • C • LateX • C++ • Java • Linux •
Languages: Bengali • English • Hindi • Spanish (Elementary) • German (Elementary)

Positions of Responsibility:

Secretary, Science Council, IISER Bhopal	2024-2025
Peer Counselor, Counselling Cell, IISER Bhopal	2023-2024
Departmental Representative, Dept. of Mathematics, IISER Bhopal	2023-2024
Math Circle Volunteer, Math Circle, IISER Bhopal.	2023-2024
Student Mentor, Student Development Council, IISER Bhopal	2023-Pres.
Quiz Club Core Committee Member	2023-Pres.
Movie Club Core Committee Member	2023-2024
Enthuzia '23 (Annual Cultural Fest) Social Media Associate	2023-2024